

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	10 Feb 2026
Team ID	LTVIP2026TMIDS36802
Project Name	Empowering India: Analysing the Evolution of Union Budget Allocations for Sustainable Growth
Maximum Marks	4 Marks

Technology Architecture:

Core Purpose: The system transforms raw Union Budget data into meaningful insights using analytical techniques.

Data Integration & Processing: Budget data is collected from government and public sources using ETL processes.

Data Storage & Performance: Processed data is stored in a structured database or data warehouse.

Analytics & Business Logic: The system analyses sector-wise allocations and year-on-year budget changes.

Visualization & User Access: Interactive dashboards present budget trends through charts and graphs.

Security & Automation: Authentication and role-based access control protect sensitive data.

Example :

A policy analyst can view year-wise Union Budget allocations and compare sector spending trends. This helps identify which sectors contribute most to sustainable economic growth

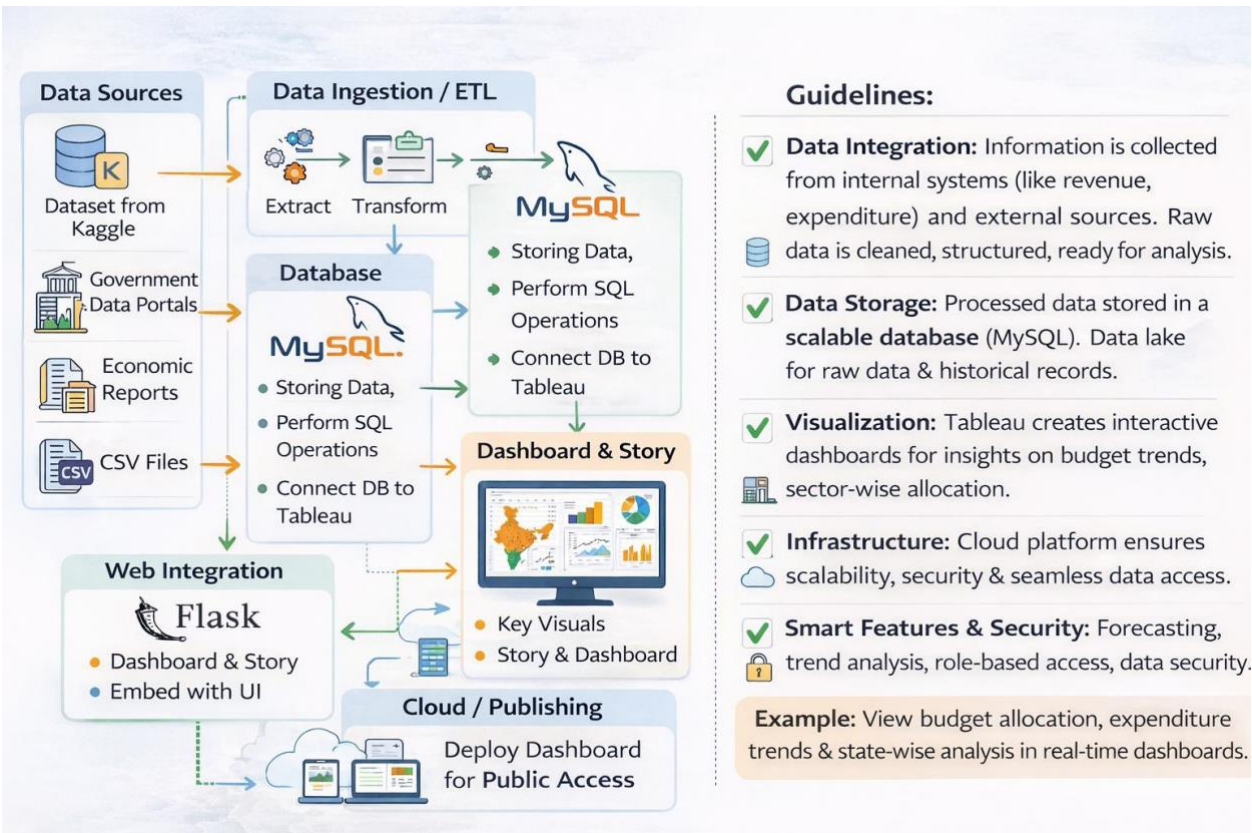


Table -1 : Components & Technologies :

S.No	Component	Description	Technology
1	User Interface	users interact with dashboards and reports for Union Budget insights	Tableau, Power BI, Web Browsers
2	Data Ingestion & ETL	Extracting budget data from government sources, cleaning, transforming, and loading it into the data storage layer	Python (Pandas, PySpark), SQL, Azure Data Factory, AWS Glue
3	Data Preparation & Analytics Engine	Data modeling, aggregations, and preparing optimized datasets for dashboards and analysis	SQL (Stored procedures, Views), Python scripts, Tableau Prep
4	Database	Central repository for structured budget and financial data	Snowflake, Google BigQuery, Amazon Redshift
5	Cloud Database	Stores raw/unstructured government data for future analysis	Amazon S3, Azure Data Lake Storage, Google Cloud Storage
6	File Storage	Temporary/archive storage for raw files, reports, or ETL outputs	Cloud Blob Storage (AWS S3, Azure Blob Storage)
7	External API-1	Integrates with government APIs to pull budget allocations, sector-wise data	REST APIs, Custom Data Connectors
8	External API-2	Connects to external datasets for economic trends, inflation, or demographic data	REST APIs, Vendor SDKs
9	Machine Learning Model	Predictive analytics for trend forecasting, allocation analysis, and anomaly detection	Python (Scikit-learn, TensorFlow), Azure ML, AWS SageMaker
10	Infrastructure (Cloud Platform)	Cloud hosting for all services and managing resources	AWS, Microsoft Azure, Google Cloud Platform
11	Infrastructure (BI Server)	Dedicated environment to host, manage, and deliver dashboards	Tableau Server / Power BI Report Server on EC2, Azure VMs, GCP.

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Tools used for processing and handling budget data	Python libraries (Pandas, PySpark), Apache Spark, SQL
2	Security Implementations	Keep sensitive financial data safe and control user access	Cloud services (AWS, Azure, GCP), Elastic data warehouses
3	Scalable Architecture	System can handle increasing users and data easily	Cloud services (AWS, Azure, GCP), Elastic data warehouses
4	Availability	Ensure dashboards and reports are always accessible	Cloud features for redundancy, load balancers, backup systems
5	Performance	Fast loading of dashboards and quick queries	Fast databases, optimized storage, efficient ETL processes